Quadratic Residues in Finite Fields and Their Extensions

Contents

1 Background: Quadratic Residues in $\mathbb{F}_p$

Let p be an odd prime. Recall that $\mathbb{F}_p^* = \mathbb{F}_p \setminus \{0\}$ is cyclic of order $p-1$.

Definition 1.1. A nonzero element $a \in \mathbb{F}_p^*$ is a *quadratic residue* (QR) if there exists $x \in \mathbb{F}_p$ such that $x^2 = a$. Otherwise, a is a *quadratic non-residue* (QNR). We set $\mathbb{F}_p^{*2} = \{a^2 : a \in \mathbb{F}_p^*\}$.

Since squaring is a 2-to-1 map on $\mathbb{F}_p^*$ (as $x^2 = (-x)^2$ and $x \neq -x$ for p odd), exactly half of the nonzero elements are QRs:

$$|\mathbb{F}_p^{*2}| = \frac{p-1}{2}.$$

Definition 1.2 (Legendre Symbol)**.** For $a \in \mathbb{Z}$ and odd prime p, the *Legendre symbol* is

$$\left(\frac{a}{p}\right) = \begin{cases} 0 & \text{if } p \mid a, \\ 1 & \text{if } a \text{ is a QR mod } p, \\ -1 & \text{if } a \text{ is a QNR mod } p. \end{cases}$$

By Euler's criterion, $\left(\frac{a}{p}\right) \equiv a^{(p-1)/2} \pmod{p}$.

2 General Finite Fields $\mathbb{F}_q$, $q = p^n$

Let $q = p^n$ for a prime p and integer $n \geq 1$. The multiplicative group $\mathbb{F}_q^*$ is cyclic of order $q - 1$.

2.1 Odd characteristic (p odd)

Proposition 2.1. *When p is odd, the squaring map $\phi : x \mapsto x^2$ is 2-to-1 on $\mathbb{F}_q^*$. Hence exactly $\frac{q-1}{2}$ elements are QRs.*

Proof. ϕ is a group homomorphism $\mathbb{F}_q^* \to \mathbb{F}_q^*$ with kernel $\{1, -1\}$, which has order 2 since p is odd. By the first isomorphism theorem, the image has index 2. □

An element $a \in \mathbb{F}_q^*$ is a QR if and only if

$$a^{(q-1)/2} = 1.$$

When $q \equiv 3 \pmod 4$, the square root of a QR a can be computed explicitly as $a^{(q+1)/4}$.

2.2 Characteristic 2 ($p = 2$)

Theorem 2.2 (All Elements are Squares in $\mathbb{F}_{2^n}$)**.** *In $\mathbb{F}_{2^n}$, every element is a perfect square. That is, the squaring map $\phi : x \mapsto x^2$ is a field automorphism.*

Proof. In characteristic 2, the *Frobenius endomorphism* $\phi(x) = x^2$ satisfies:

$$\begin{aligned} \phi(x + y) &= (x + y)^2 = x^2 + 2xy + y^2 = x^2 + y^2 = \phi(x) + \phi(y), \\ \phi(xy) &= (xy)^2 = x^2y^2 = \phi(x)\phi(y), \end{aligned}$$

so ϕ is a ring homomorphism. Since $\mathbb{F}_{2^n}$ is a field, $\ker \phi = \{0\}$, making ϕ injective. A finite injective map is bijective, so ϕ is a field automorphism. In particular, every element has a unique square root. □

Corollary 2.3. *The notion of a quadratic non-residue is vacuous in $\mathbb{F}_{2^n}$: every element is a QR.*

Explicit Square Root Formula

For any $a \in \mathbb{F}_{2^n}$, the unique square root is

$$\sqrt{a} = a^{2^{n-1}}.$$

Proof. $\left(a^{2^{n-1}}\right)^2 = a^{2^n}$. By Fermat's little theorem applied to $\mathbb{F}_{2^n}$, we have $a^{2^n} = a$ for all $a \in \mathbb{F}_{2^n}$. Hence $\left(a^{2^{n-1}}\right)^2 = a$. □

3 Quadratic Residues in Extension Fields

A fundamental phenomenon is that non-residues in a base field can become residues after passing to an extension.

Theorem 3.1. *Let p be an odd prime and $q = p^n$. Every element of $\mathbb{F}_q$ is a quadratic residue in $\mathbb{F}_{q^2}$.*

Proof. Let $a \in \mathbb{F}_q^*$. An element $b \in \mathbb{F}_{q^2}^*$ is a QR in $\mathbb{F}_{q^2}$ if and only if

$$b^{(q^2-1)/2} = 1.$$

We compute:

$$a^{(q^2-1)/2} = a^{(q-1)(q+1)/2} = \left(a^{q-1}\right)^{(q+1)/2}.$$

Since $a \in \mathbb{F}_q^*$, Fermat's little theorem gives $a^{q-1} = 1$. Therefore

$$a^{(q^2-1)/2} = 1^{(q+1)/2} = 1,$$

so a is a QR in $\mathbb{F}_{q^2}$. □

Remark 3.2. This shows that the degree-2 extension "resolves" all non-residues from the base field. However, $\mathbb{F}_{q^2}$ itself still contains QNRs (about half of its own nonzero elements are non-residues with respect to $\mathbb{F}_{q^2}$).

Corollary 3.3. *Every QNR of $\mathbb{F}_p$ is a QR in $\mathbb{F}_{p^2}$.*

More generally, we have:

Proposition 3.4. *Let $a \in \mathbb{F}_q^*$ and let $\mathbb{F}_{q^m}$ be an extension of degree m. Then a is a QR in $\mathbb{F}_{q^m}$ if and only if*

$$a^{(q^m-1)/2} = 1.$$

In particular, if m is even, then $a^{q-1} = 1$ implies $a^{(q^m-1)/2} = 1$, so every element of $\mathbb{F}_q^*$ is a QR in any even-degree extension.

4 Summary Table

Field / Setting	**Squaring map**	**Number of QRs (nonzero)**
$\mathbb{F}_p$, p odd	2-to-1	$(p-1)/2$
$\mathbb{F}_{p^n}$, p odd	2-to-1	$(p^n-1)/2$
$\mathbb{F}_{2^n}$	Bijection (automorphism)	$2^n - 1$ (all)
$\mathbb{F}_q \hookrightarrow \mathbb{F}_{q^2}$, q odd	—	All of $\mathbb{F}_q^*$ become QRs
$\mathbb{F}_q \hookrightarrow \mathbb{F}_{q^m}$, m even	—	All of $\mathbb{F}_q^*$ become QRs

5 Cryptographic Implications

The structural difference between odd and even characteristic has significant consequences for cryptography.

- **QR-based cryptography** (e.g., Goldwasser–Micali encryption, Blum–Blum–Shub PRG) relies on the computational hardness of distinguishing QRs from QNRs. This hardness disappears entirely in characteristic 2, since there are no non-residues.
- **Square roots in** $\mathbb{F}_{2^n}$ are cheap: computing $a^{2^{n-1}}$ costs only $n-1$ squarings (each a Frobenius application), far cheaper than in $\mathbb{F}_p$.
- **Pairing-based cryptography** often works over extensions $\mathbb{F}_{p^k}$ of prime fields. Theorem 3.1 explains why certain subgroup elements (living in $\mathbb{F}_p$) are automatically squares in the full extension group.
- **Elliptic curve point compression** over $\mathbb{F}_p$ uses the Legendre symbol to recover the correct y-coordinate from $y^2 = x^3 + ax + b$. Over $\mathbb{F}_{2^n}$, a completely different technique is needed since all elements are squares.